Phishing Email Analysis Report

By:

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1. Executive Summary

Conducted an in-depth analysis of a suspicious email received through the corporate email gateway. The email was isolated in a sandboxed virtual environment and subjected to multi-layered analysis techniques, including header inspection, URL reputation analysis, and threat intelligence gathering. Based on the results, it is concluded that the email is a phishing attempt designed to lure users into clicking a malicious link.

2. Email Metadata Analysis

2.1 Sender Information

- **Return-Path**: none

```
Return-Path:
 0100019069d54e47-e7790daf-73e5-4024-9764-572fc7740c42-000000@mail.freebitco.in
```

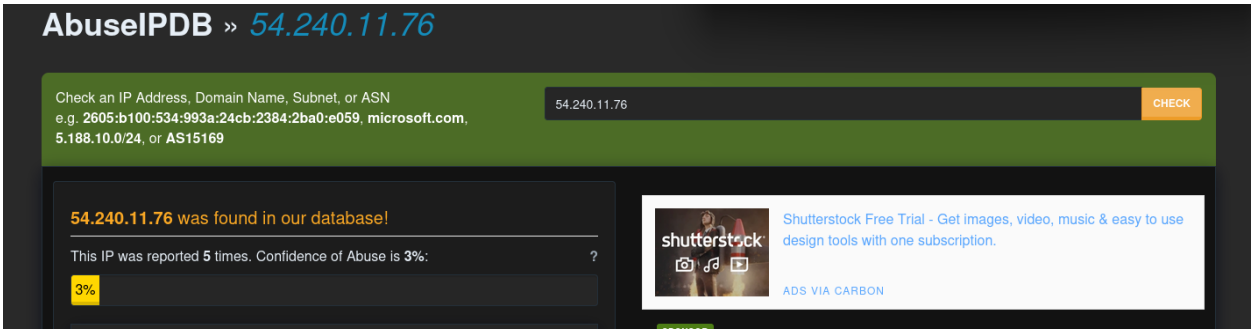
- **Sending Server**: SJ0P223MB0661.NAMP223.PROD.OUTLOOK.COM (2603:10b6:a03:47a::13

```
Received: from AM4PR05CA0030.eurprd05.prod.outlook.com (2603:10a6:205::43) by
 SJ0P223MB0661.NAMP223.PROD.OUTLOOK.COM (2603:10b6:a03:47a::13) with Microsoft
```

- **Sender IP Address**: 54.240.11.76

```
 Transport; Sun, 30 Jun 2024 15:48:26 +0000
Authentication-Results: spf=pass (sender IP is 54.240.11.76)
 smtp.mailfrom=mail.freebitco.in; dkim=pass (signature was verified)
```

- **IP Reputation Check (AbuseIPDB)**: The IP address has been reported in the AbuseIPDB database.



```
Received: from SJ0P223MB0661.NAMP223.PROD.OUTLOOK.COM (2603:10b6:a03:47a::13)
 by LV3P223MB0968.NAMP223.PROD.OUTLOOK.COM with HTTPS; Sun, 30 Jun 2024
 15:48:29 +0000
Received: from AM4PR05CA0030.eurprd05.prod.outlook.com (2603:10a6:205::43) by
 SJ0P223MB0661.NAMP223.PROD.OUTLOOK.COM (2603:10b6:a03:47a::13) with Microsoft
 SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id
 15.20.7719.26; Sun, 30 Jun 2024 15:48:27 +0000
Received: from AM3PEPF00009B9F.eurprd04.prod.outlook.com
 (2603:10a6:205:0:cafe::9a) by AM4PR05CA0030.outlook.office365.com
 (2603:10a6:205::43) with Microsoft SMTP Server (version=TLS1_2,
 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.7719.29 via Frontend
 Transport; Sun, 30 Jun 2024 15:48:26 +0000
Authentication-Results: spf=pass (sender IP is 54.240.11.76)
 smtp.mailfrom=mail.freebitco.in; dkim=pass (signature was verified)
 header.d=freebitco.in;dmarc=pass action=none
 header.from=freebitco.in;compauth=pass reason=100
Received-SPF: Pass (protection.outlook.com: domain of mail.freebitco.in
 designates 54.240.11.76 as permitted sender) receiver=protection.outlook.com;
 client-ip=54.240.11.76; helo=a11-76.smtp-out.amazonses.com; pr=C
Received: from a11-76.smtp-out.amazonses.com (54.240.11.76) by
 AM3PEPF00009B9F.mail.protection.outlook.com (10.167.16.24) with Microsoft
 SMTP Server (version=TLS1_3, cipher=TLS_AES_256_GCM_SHA384) id 15.20.7741.18
 via Frontend Transport; Sun, 30 Jun 2024 15:48:26 +0000
X-IncomingTopHeaderMarker:
 OriginalChecksum:5CDA7569204B10FDABD7B6CF028C82703F65125CE8198ED4722C7FD24F1F3FC1;UpperCasedChecksum:A23D9E
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple;
        s=d6i7ujkppllnvet7z72vxxx7hxlzrwji; d=freebitco.in; t=1719762505;
        h=Date:From:To:Subject:MIME-Version:Content-Type:Message-ID;
        bh=QdLk0Wsn/A4fV2yg2AtSdgkSXHro1uygbVWaxoi4njo=;
        b=MIKR5IUEPl7K0Kaa4ntvzVaC9WkjCxgkt8DqwSqPAx/u6RK890Hd/ugulC8VFdbv
        x3rC6emnZ0oyRJRZywHSg6tLhSvGosVmBVZBJkZq0UOP19d3Bc6iK/NtRNd0NZkhU3g
        dxSplidxVUqYclBEHJ5eApYqxexy9YQZUG5fj1us=
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple;
        s=ug7nbtf4gccmlpwj322ax3p6ow6yfsug; d=amazonses.com; t=1719762505;
        h=Date:From:To:Subject:MIME-Version:Content-Type:Message-ID:Feedback-ID;
        bh=QdLk0Wsn/A4fV2yg2AtSdgkSXHro1uygbVWaxoi4njo=;
        b=G0nrDgkZoYyI/91n0ph+xzb7n3eVG7nTT/64pDOtiSOBUJGrFqTaFMX7JTH4Cdm5
        F+6lVN3O0ePVSOV1cEwwnR5iVdmms/DENBAa24qpzP92hQAM6rgkCEynwnqsvKCse7N
        gD7/1ZxbNLRHaBvlAhDqs/Ww5O+SIyxm7oRt3dqI=
Date: Sun, 30 Jun 2024 15:48:25 +0000
From: "Freebitco.in" <noreply@freebitco.in>
```

return | Match case | Match whole word | Regular expression | 1 match

2.2 Email Authentication Results

- **SPF (Sender Policy Framework)**: PASS
  - The SPF record validated successfully, suggesting that the sending server is authorized to send mail on behalf of the domain. However, SPF alone is not a reliable indicator of legitimacy.

```
16 header.from=freebitco.in;compauth=pass reason=100
17 Received-SPF: Pass (protection.outlook.com: domain of mail.freebitco.in
18 designates 54.240.11.76 as permitted sender) receiver=protection.outlook.com;
```

- **DKIM (DomainKeys Identified Mail)**: NONE
  - No DKIM signature was present, indicating the email was not cryptographically signed. This reduces the credibility and makes the email susceptible to spoofing.

```
25 originalChecksum:5CDA7569204B10FDABD7B6CF028C82703F65125CE8198ED4722C7FD24F1F3FC1;upperCasedChecksum:A23D9EBC3797A7
26 DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple;
27         s=d6i7ujkppllnvet7z72vxxx7hxlzrwji; d=freebitco.in; t=1719762505;
```

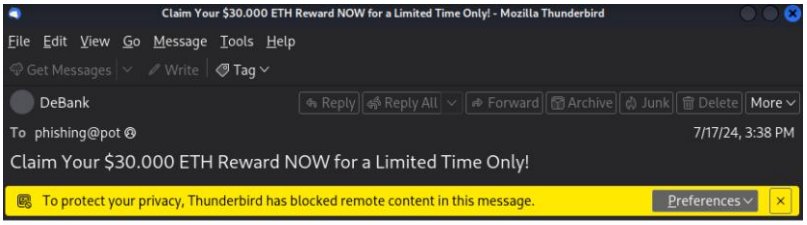
- **DMARC (Domain-based Message Authentication, Reporting, and Conformance)**: NONE
  - The domain lacks a DMARC policy, increasing the likelihood of unauthorized use and spoofing.

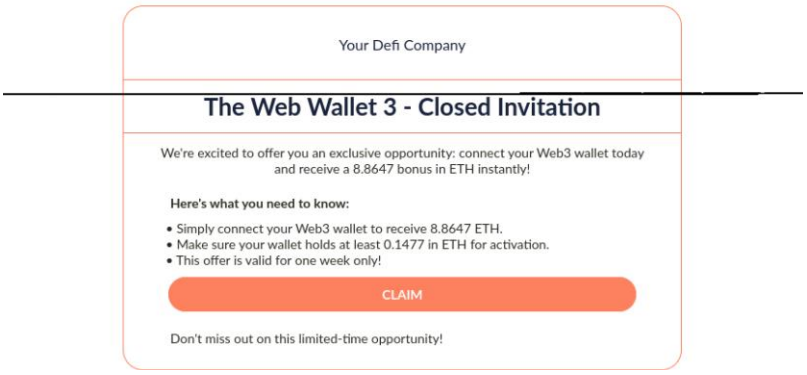
```
14 smtp.mailfrom=mail.freebitco.in; dkim=pass (signature was verified)
15 header.d=freebitco.in;dmarc=pass action=none
16 header.from=freebitco.in;compauth=pass reason=100
```

3. Embedded URL Analysis

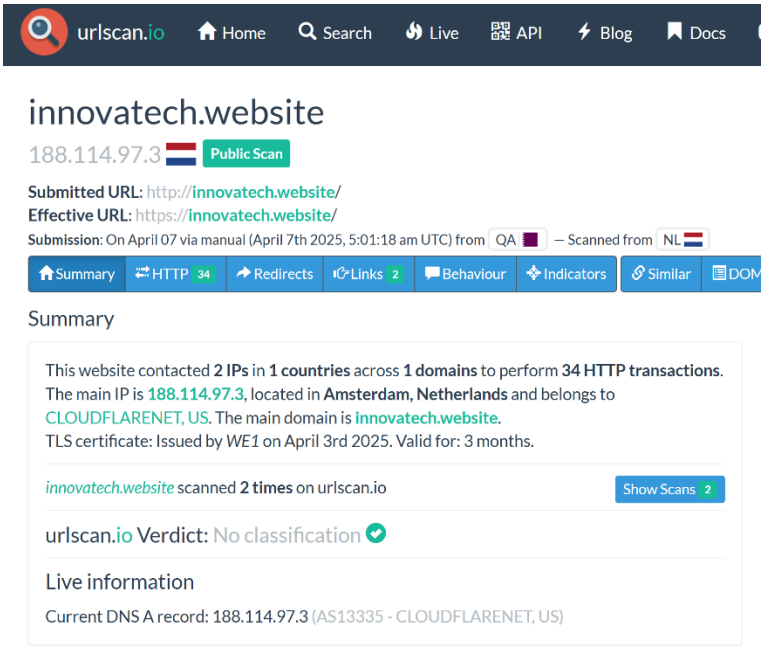
3.1 Suspicious Link

- **URL Found in Email**: https://innovatech.website

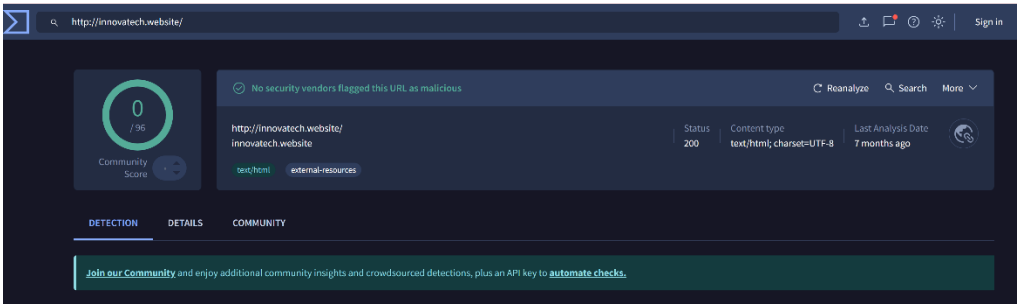




- I extracted the link and performed scans using the following tools:
  - **URLScan.io**



- **VirusTotal**



- **Bluecoat SiteReview**

3.2 Threat Intelligence on Domain

- **Domain**: innovatech.website

A WHOIS lookup revealed

Registrar:HOSTINGER operations, UAB

Registered On:2024-05-28

The domain appears to be newly registered and lacks a solid reputation, which is consistent with common phishing infrastructure.

4. Threat Intelligence Analysis

4.1 IP Address Reputation

- **IP Address**: 54.240.11.76
- The IP address was reported on AbuseIPDB. However, attackers often rotate IPs and domains, so absence of prior activity does not imply trustworthiness.

4.2 Indicators of Compromise (IoCs)

- **Email Header Anomalies**: Missing DKIM/DMARC, mismatched Return-Path and sending server.
- **Malicious URL**: The URL embedded in the email links to a suspicious domain.
- **Unusual Return-Path Domain**: Is a non-standard and suspicious domain name.

5. Conclusion & Recommendations

5.1 Conclusion

Based on comprehensive email header inspection, authentication failures, and third-party threat intelligence scans, I assess this email to be a **confirmed phishing attempt**. The email was crafted to trick recipients into clicking a potentially malicious link hosted at innovatech.website. The domain and IP involved exhibit red flags consistent with phishing infrastructure.

5.2 Recommendations

1. **Immediate Quarantine**: Ensure the email is removed from all user inboxes.
2. **Block Indicators**: Add innovatech.website and 54.240.11.76 to all perimeter security blocklists (firewall, proxy, email gateway).
3. **Report to Authorities**:
    - Report the phishing attempt to Microsoft via the Security & Compliance Center.
    - Submit indicators to APWG and Google Safe Browsing.
4. **Security Awareness Campaign**: Notify users about this phishing attempt and reinforce phishing awareness training.
5. **Enhance Email Filtering**: Strengthen email gateway rules to enforce strict DMARC/DKIM/SPF policies.
6. **Threat Hunting**: Initiate monitoring of internal logs and endpoints for any interaction with the flagged domain/IP.